Gargi Mitra

Security Researcher, PhD | AI Security, AI for Security, Network and Systems Security

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gargi-mitra.github.io/website | github.com/ICS-Sniper | Google Scholar

Summary

Security researcher (PhD, IIT Madras; Postdoctoral Research Fellow, University of British Columbia, 2023–Aug 2026) who identifies the security assumptions that break when cloud connectivity, encryption, or AI/ML is layered onto legacy infrastructure, proves them with proof-of-concept attacks, and designs detection and mitigation strategies. Work spans encrypted traffic analysis, ICS/OT security, adversarial machine learning and LLM-based security tooling, with publications at NDSS, EuroSys, IEEE/IFIP DSN, IEEE INFOCOM and IEEE TDSC. Led a five-member, multi-year project and co-authored its CAD 500K grant; research covered by WIRED, with a mitigation adopted by Netflix.

Skills

AI security and security of AI: Adversarial machine learning; evasion attacks and defenses; adversarial input generation (URET); inference-time false data injection; ML robustness; AI red-teaming tooling; AI threat modeling (MITRE ATT&CK and ATLAS); ML-enabled medical devices

AI for security: Large language model (LLM) pipelines with retrieval-augmented generation (RAG); prompt engineering; LLM output evaluation; automated vulnerability discovery (CVE, NVD); attack-path generation; agentic AI for detection

Network security and detection: Encrypted traffic analysis (TLS 1.3, HTTP/2, VPN: WireGuard and OpenVPN); traffic fingerprinting; side-channel analysis; network intrusion detection (Zeek, Kitsune); anomaly detection (network traffic and system log-based); IDS evasion analysis; detection engineering; MITRE ATT&CK mapping; coverage and false-positive trade-offs; pcap analysis; TCP/IP

Systems, ICS/OT and IoT security: Vulnerability research on protocol- and design-level weaknesses; root-cause and exploitability analysis; proof-of-concept attack development; on-path (MITM) attacks; targeted denial of service; cloud-connected ICS and SCADA; Modbus/TCP; CIP and EtherNet/IP; Allen-Bradley PLCs; IoT privacy; information flow control; NIST SP 800-82; IEC 62443

Threat modeling: STPA-Sec; FDA-cleared medical device cybersecurity; GDPR

Programming and tools: Python; C; Bash; PyTorch; scikit-learn; Scapy; pyshark; NetfilterQueue; Zeek scripting; Wireshark and tshark; Docker; Git; Linux; kernel networking (SCION)

Experience

Postdoctoral Research Fellow, Electrical and Computer Engineering | **Jan 2023 – Aug 2026**
The University of British Columbia, Vancouver, BC, Canada

Network and ICS/OT security: ICS-Sniper | ics-sniper.github.io/ics-sec-website

- Conceived and led **ICS-Sniper** (NDSS 2027), the first targeted blackhole attack on encrypted ICS/OT traffic: an on-path adversary uses VPN traffic metadata (packet sizes, direction, timing) to drop a few packets carrying critical control data, disrupting the physical process without breaking encryption and while evading state-of-the-art ICS attack detectors.
- Built the analysis pipeline: feature extraction from TLS- and VPN-encapsulated SCADA traffic, recovery of control-loop periodicity, clustering to identify target flows, and selective packet dropping in live traffic.
- Built the first realistic testbeds for network attacks on cloud-connected ICS: hardware and software Secure Water Treatment testbeds (H-SWaT, S-SWaT) with Allen-Bradley PLCs. Demonstrated three attacks, evaluated network-layer mitigations such as traffic shaping, and open-sourced the attack code, testbed code and detector baselines.
- Led the five-member, multi-year project; co-authored the National Cybersecurity Consortium grant funded at CAD 500K (2024); planned milestones, trained students and interns, and reported progress to the consortium quarterly.

Security of AI: ML-enabled medical devices

- Showed that integrating AI/ML with legacy medical devices opens new inference-time attack paths, by injecting adversarial inputs into an ML-enabled blood glucose monitoring system through a known Bluetooth vulnerability (IEEE/ACM CHASE 2024).
- Built LLM- and RAG-based security tools: **SAM** automates STPA-Sec threat analysis, using vulnerability databases and LLMs to generate attack paths at design time (ACM CCS 2024 HealthSec workshop); **MedAIScout** retrieves known ML vulnerabilities for red teaming, finding relevant ones in 4 of 5 FDA-approved devices (NeurIPS 2024 Red Teaming GenAI workshop; 2nd prize, UBC Ideas with Impact).
- Improved detection of evasion attacks on DNNs by training anomaly detectors selectively with risk profiling, raising recall by up to 27.5% (IEEE/IFIP DSN 2025 DSML workshop); a follow-up on outlier exposure is under review.

- Presented findings to the FDA and to security practitioners at Medtronic, GE and Medcrypt.

Systems security, leadership and mentoring

- Co-authored **Turnstile** (EuroSys 2026), a hybrid information flow control framework for GDPR-compliant IoT applications, and **ImmunoPlane** (IoTDI 2024), middleware that brings adaptivity to distributed IoT applications.
- Co-supervised 1 PhD and 2 MASc students and co-mentored 10 undergraduate researchers; contributed technical and progress reports to projects funded by the National Research Council and the Department of National Defence of Canada.

Mitacs Globalink Research Intern | **Sep 2021 – Dec 2021**
The University of British Columbia (with IIT Madras), Vancouver, BC, Canada
Encrypted traffic analysis-based attacks on Internet-connected industrial control systems.

PhD Researcher, Computer Science and Engineering | **2015 – 2022**
Indian Institute of Technology Madras, Chennai, India

- Designed the first traffic-analysis attack that defeats HTTP/2 multiplexing, forcing the server to serialize responses and breaking the privacy of a real-world HTTP/2 website 90% of the time (IEEE/IFIP DSN 2020).
- Built **Snoopy**, a webpage fingerprinting framework for mass surveillance within website query limits (finite query model): about 90% accuracy on most websites, about 97% in an ensemble with ML (IEEE TDSC 2022).
- Inferred viewers' choices in Netflix's interactive "Black Mirror: Bandersnatch" from encrypted traffic with a worst-case accuracy of 96% (**White Mirror**, ACM SIGCOMM 2019 poster); covered by WIRED, Financial Express and Gadgets 360; Netflix adopted one of the proposed mitigations after responsible disclosure.
- Designed tamper-resistant QoS measurement for untrusted multi-vendor service function chains (IEEE INFOCOM 2019); worked with Adrian Perrig's group (ETH Zurich) on a kernel implementation of SCION.

Programmer Analyst | **2013 – 2015**
Cognizant Technology Solutions, India
Data warehousing for enterprise data systems within production software development workflows.

Independent Projects (in progress)

- **Detection characteristics framework:** uses AI to translate attack tactics and techniques from MITRE ATT&CK and ATLAS into queryable network characteristics, specifying what each detection observes.
- **Zeek and Kitsune network IDS:** a live pipeline with Zeek for capture and features and Kitsune's autoencoder ensemble as the detector.

Education

PhD (MS + PhD Dual Degree), Computer Science and Engineering | **2015 – 2022**
Indian Institute of Technology Madras, Chennai, India
Dissertation: Encrypted Traffic Analysis-Based Cyber Surveillance and Possible Mitigation Techniques

BTech, Computer Science and Engineering | **2009 – 2013**
St. Thomas' College of Engineering & Technology (MAKAUT), Kolkata, India

Selected Publications

- **Network Security and Privacy (First author):** *ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic* (NDSS 2027); *Snoopy: A Webpage Fingerprinting Framework with Finite Query Model for Mass-Surveillance* (IEEE TDSC 2022); *Depending on HTTP/2 for Privacy? Good Luck!* (IEEE/IFIP DSN 2020); *White Mirror* (ACM SIGCOMM 2019 poster).
- **AI security:** *Systems-Theoretic and Data-Driven Security Analysis in ML-enabled Medical Devices (First author)* (Springer CCIS 2026, invited); *MedAIScout: Automated Retrieval of Known Machine Learning Vulnerabilities in Medical Applications* (NeurIPS 2024 Red Teaming GenAI workshop). Other papers are cited above; full list on Google Scholar.

Awards, Service and Talks

- ACM CCS Distinguished Reviewer Award (2024; top 10% of 527 PC members); program committees of ACM CCS 2024–2026, ACM AsiaCCS 2026 and IEEE CSR 2025.
- First prize, CSAW India (NYU and IIT Kanpur): Applied Research Competition (2020) and Embedded Security Challenge, for defending PLC-based industrial control systems (2017); Mitacs Globalink Research Award (2021).
- Invited talks at Idaho National Laboratory (2024) and ETS Montreal (2026); organized Birds-of-a-Feather sessions at USENIX Security 2023 and 2025 and a summer workshop on AI security at UBC (2025); participant, AI safety and security workshop at Microsoft, Redmond (2026).